Heart Disease Prediction

Overview

This repository contains a project focused on heart disease prediction. The data, derived from heart patients, includes various health metrics such as age, blood pressure, heart rate, and more. The primary objective is to create a predictive model that accurately identifies individuals at risk of heart disease.

Problem

In this project, we delvelop into a dataset encapsulating various health metrics from heart patients, including age, blood pressure, heart rate, and more. Our goal is to develop a predictive model capable of accurately identifying individuals with heart disease. Given the grave implications of missing a positive diagnosis, our primary emphasis is on ensuring that the model identifies all potential patients, making recall for the positive class a crucial metric.

Objectives

The objectives of the project are as follows:

1. **Data collection**: collect the dataset from Kaggle platform
2. The dataset comprises various metrics related to heart health. The features of the dataset are described in the table below:

Variable Name	Description
age	Age of the patient in years
sex	Gender of the patient (0 = male, 1 = female)
cp	Chest pain type: 0: Typical angina 1: Atypical angina 2: Non-anginal pain 3: Asymptomatic
trestbps	Resting blood pressure in mm Hg
Chol	Serum cholesterol in mg/dl
fbs	Fasting blood sugar level
resting	Resting electrocardiographic results: 0: Normal

Variable Name	Description
	1: Having ST-T wave abnormality 2: Showing probable or definite left ventricular hypertrophy
thalach	Maximum heart rate achieved during a stress test
exang	Exercise-induced angina (1 = yes, 0 = no)
old peak	ST depression induced by exercise relative to rest
slope	Slope of the peak exercise ST segment: 0: Upsloping 1: Flat 2: Downs loping
ca	Number of major vessels (0-4) coloured by fluoroscopy
thal	Thallium stress test result: 0: Normal 1: Fixed defect 2: Reversible defect 3: Not described
target	Heart disease status (0 = no disease, 1 = presence of disease)

3. **Exploratory Data Analysis :**
    - Univariate Analysis
    - Bivariate Analysis
4. **Data Preprocessing**: Prepare the data for future machine learning tasks.
    - Remove irrelevant features
    - Address missing values
    - Check the null values
5. **Model Building**: Develop and refine the prediction models.
    - Establish pipelines for models that require scaling
    - Implement and tune classification models including KNN, SVM, Decision Tree, and Random Forest

- Emphasize achieving high recall for class 1, ensuring comprehensive identification of heart patients

6. **Evaluate and Compare Model Performance**: compare with each and every algorithm by using the accuracy value

Machine Learning Models

1. Logistic Regression

A linear classification algorithm used as one of the baseline models.

It estimates the probability of the target belonging to a particular class and is commonly used for binary classification problems.

2. Random Forest

An ensemble learning algorithm that combines multiple decision trees to produce a more robust prediction.

Random Forest can capture nonlinear relationships between features and the target variable.

3. Naive Bayes

A probabilistic classification algorithm based on Bayes' theorem.

It assumes conditional independence between features and provides a simple and efficient classification approach.

4. K-Nearest Neighbors

KNN predicts the class of a data point based on the classes of its nearest neighbors.

The algorithm is for understanding distance-based classification.

5. Decision Tree

A tree-based classification algorithm that makes predictions by recursively splitting the .

dataset-based **Model Comparison**

6.SelectKbest:

The dataset contains 14 features, analyse the data which is performed well that features used for the model (6 seleckbest features)

The main objective of the project is not simply to train one model, but to understand how different classification algorithms perform on the same dataset.

Model	**Type**
Logistic Regression	Linear classifier
Random Forest	Ensemble / tree-based

Model	Type
Naive Bayes	Probabilistic classifier
KNN	Distance-based
Decision Tree	Tree-based
SelectKBest	Best features

This comparison provides a useful introduction to the strengths and weaknesses of different supervised learning approaches.

on feature values.

The project also explores adjustments to the algorithm to improve predictive performance.

Machine Learning Workflow

The project follows a standard supervised machine learning workflow:

1. Data Collection

Load the heart disease dataset from heart.csv.

2. Data Exploration

Explore:

- Dataset dimensions
- Feature distributions
- Target distribution
- Statistical properties
- Relationships between variables

3. Data Preprocessing

Prepare the dataset for machine learning by processing the relevant features and target values.

4. Model Training

Train multiple classification algorithms:

Logistic Regression

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Random Forest

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Naive Bayes

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KNN

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Decision Tree

5. Model Evaluation

Compare the models using classification performance metrics and identify promising approaches.

6. Model Improvement

The project experiments with different aspects of the algorithms to improve predictive accuracy.

Technologies

The project primarily uses the Python data science and machine learning ecosystem.

The project also explores adjustments to the algorithm to improve predictive performance.

- Relationships between variables

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Technologies

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Technology	**Purpose**
Python	Programming language
Jupyter Notebook	Interactive experimentation
Pandas	Data manipulation
NumPy	Numerical computation
Matplotlib	Data visualization
Seaborn	Statistical visualization

Results

The project ultimately explores improvements to the classification models, including experimentation with the Logistic Regression approach

In this project, selectkbest features used and we've built a Logistic Regression model that predicts heart disease with approximately 81% accuracy.

Thank you